PREMIUM

Full-Length Mock Test

Mock 03

Civil Engineering

GATE CE

Total Questions	60
Maximum Marks	108
Duration	180 minutes (3 Hours)
Exam Pattern	MCQ + MSQ + NAT
Negative Marking	MCQ/MSQ: -0.33 per wrong answer
Designed For	EduNeuro Premium

Easy 20% | Moderate 33% | Hard 47%

Based on Previous Year Question Pattern Analysis | Generated by EduNeuro Premium

General Instructions

This question paper contains THREE sections: Section A, Section B, and Section C.

Section A: Multiple Choice Questions (MCQ) — carries one or two marks each. Only ONE option is correct.

Section B: Multiple Select Questions (MSQ) — carries two marks each. ONE OR MORE than one option(s) is/are correct.

Section C: Numerical Answer Type (NAT) — carries one or two marks. Enter the numerical value in the space provided.

Negative Marking: -0.33 marks will be deducted for each wrong MCQ and each wrong MSQ answer.

No negative marking for NAT questions.

There are 60 questions for a total of 100 marks.

Duration: 3 hours (180 minutes).

Use the space provided for rough work. Scribbling on the question paper is permitted.

Do not ask for clarification from the invigilator. Interpret questions as given.

Paper Structure

Section	Type	Questions	Marks/Q	Total Marks	Negative
A	MCQ	35	1 / 2	55	-0.33
B	MSQ	5	2	10	-0.33
C	NAT	20	1 / 2	35	None
Total		**60**		**100**	

Marking Scheme

- Correct MCQ: +1 mark (or +2 marks as specified per question)
- Wrong MCQ: -0.33 marks
- Correct MSQ: +2 marks
- Wrong MSQ (partial or no match): -0.33 marks
- Correct NAT: +1 mark (or +2 marks as specified per question)
- No answer / Unattempted: 0 marks
- NAT wrong answer: 0 marks (no negative marking)

Section A — Multiple Choice Questions (MCQ)

Q.1 [1 Mark] [EASY] [Structural Engineering — Deflection]

Max BM in simply supported beam with UDL is at:

A) Supports
B) Midspan
C) Quarter
D) Anywhere

Q.2 [1 Mark] [EASY] [Geotechnical Engineering — Soil Properties]

Terzaghi's bearing capacity assumes:

A) General shear
B) Local shear
C) Punching shear
D) None

Q.3 [1 Mark] [EASY] [Transportation Engineering — Traffic Engineering]

Bernoulli's equation is based on:

A) Mass conservation
B) Energy conservation
C) Momentum conservation
D) Force conservation

Q.4 [1 Mark] [EASY] [Water Resources — Dams]

Prandtl's mixing length theory is used for:

A) Laminar flow
B) Turbulent flow
C) Compressible flow
D) Open channel

Q.5 [1 Mark] [EASY] [Environmental Engineering — Water Treatment]

CPM stands for:

A) Critical Path Method
B) Cost Per Mile
C) Continuous Process Model
D) Critical Process Method

Q.6 [1 Mark] [EASY] [Construction Management — Project Planning]

Max BM in simply supported beam with UDL is at:

A) Supports
B) Midspan
C) Quarter
D) Anywhere

Q.7 [1 Mark] [EASY] [Structural Engineering — Influence Lines]

Terzaghi's bearing capacity assumes:

A) General shear

B) Local shear

C) Punching shear

D) None

Q.8 [1 Mark] [EASY] [Geotechnical Engineering — Slope Stability]

Bernoulli's equation is based on:

A) Mass conservation
B) Energy conservation
C) Momentum conservation
D) Force conservation

Q.9 [1 Mark] [EASY] [Transportation Engineering — Pavement Design]

Prandtl's mixing length theory is used for:

A) Laminar flow
B) Turbulent flow
C) Compressible flow
D) Open channel

Q.10 [1 Mark] [EASY] [Water Resources — Hydrology]

CPM stands for:

A) Critical Path Method
B) Cost Per Mile
C) Continuous Process Model
D) Critical Process Method

Q.11 [1 Mark] [EASY] [Environmental Engineering — Noise Pollution]

Max BM in simply supported beam with UDL is at:

A) Supports
B) Midspan
C) Quarter
D) Anywhere

Q.12 [1 Mark] [EASY] [Construction Management — CPM/PERT]

Terzaghi's bearing capacity assumes:

A) General shear
B) Local shear
C) Punching shear
D) None

Q.13 [2 Marks] [MODERATE] [Structural Engineering — Deflection]

Bernoulli's equation is based on:

A) Mass conservation
B) Energy conservation
C) Momentum conservation
D) Force conservation

Q.14 [2 Marks] [MODERATE] [Geotechnical Engineering — Slope Stability]

Prandtl's mixing length theory is used for:

A) Laminar flow

B) Turbulent flow

C) Compressible flow

D) Open channel

Q.15 [2 Marks] [MODERATE] [Transportation Engineering — Traffic Engineering]

CPM stands for:

A) Critical Path Method
B) Cost Per Mile
C) Continuous Process Model
D) Critical Process Method

Q.16 [2 Marks] [MODERATE] [Water Resources — Dams]

Max BM in simply supported beam with UDL is at:

A) Supports
B) Midspan
C) Quarter
D) Anywhere

Q.17 [2 Marks] [MODERATE] [Environmental Engineering — Sewage Treatment]

Terzaghi's bearing capacity assumes:

A) General shear
B) Local shear
C) Punching shear
D) None

Q.18 [2 Marks] [MODERATE] [Construction Management — Contracts]

Bernoulli's equation is based on:

A) Mass conservation
B) Energy conservation
C) Momentum conservation
D) Force conservation

Q.19 [2 Marks] [MODERATE] [Structural Engineering — Deflection]

Prandtl's mixing length theory is used for:

A) Laminar flow
B) Turbulent flow
C) Compressible flow
D) Open channel

Q.20 [2 Marks] [MODERATE] [Geotechnical Engineering — Slope Stability]

CPM stands for:

A) Critical Path Method
B) Cost Per Mile
C) Continuous Process Model
D) Critical Process Method

Q.21 [2 Marks] [MODERATE] [Transportation Engineering — Pavement Design]

Max BM in simply supported beam with UDL is at:

A) Supports

B) Midspan

C) Quarter

D) Anywhere

Q.22 [2 Marks] [MODERATE] [Water Resources — Hydrology]

Terzaghi's bearing capacity assumes:

A) General shear
B) Local shear
C) Punching shear
D) None

Q.23 [2 Marks] [MODERATE] [Environmental Engineering — Noise Pollution]

Bernoulli's equation is based on:

A) Mass conservation
B) Energy conservation
C) Momentum conservation
D) Force conservation

Q.24 [2 Marks] [MODERATE] [Construction Management — Valuation]

Prandtl's mixing length theory is used for:

A) Laminar flow
B) Turbulent flow
C) Compressible flow
D) Open channel

Q.25 [2 Marks] [MODERATE] [Structural Engineering — Moment Distribution]

CPM stands for:

A) Critical Path Method
B) Cost Per Mile
C) Continuous Process Model
D) Critical Process Method

Q.26 [2 Marks] [MODERATE] [Geotechnical Engineering — Slope Stability]

Max BM in simply supported beam with UDL is at:

A) Supports
B) Midspan
C) Quarter
D) Anywhere

Q.27 [2 Marks] [MODERATE] [Transportation Engineering — Traffic Engineering]

Terzaghi's bearing capacity assumes:

A) General shear
B) Local shear
C) Punching shear
D) None

Q.28 [2 Marks] [MODERATE] [Water Resources — Irrigation]

Bernoulli's equation is based on:

A) Mass conservation

B) Energy conservation

C) Momentum conservation

D) Force conservation

Q.29 [2 Marks] [MODERATE] [Environmental Engineering — Noise Pollution]

Prandtl's mixing length theory is used for:

A) Laminar flow
B) Turbulent flow
C) Compressible flow
D) Open channel

Q.30 [2 Marks] [MODERATE] [Construction Management — Quality Control]

CPM stands for:

A) Critical Path Method
B) Cost Per Mile
C) Continuous Process Model
D) Critical Process Method

Q.31 [2 Marks] [MODERATE] [Structural Engineering — Deflection]

Max BM in simply supported beam with UDL is at:

A) Supports
B) Midspan
C) Quarter
D) Anywhere

Q.32 [2 Marks] [MODERATE] [Geotechnical Engineering — Slope Stability]

Terzaghi's bearing capacity assumes:

A) General shear
B) Local shear
C) Punching shear
D) None

Q.33 [2 Marks] [HARD] [Transportation Engineering — Traffic Engineering]

Bernoulli's equation is based on:

A) Mass conservation
B) Energy conservation
C) Momentum conservation
D) Force conservation

Q.34 [2 Marks] [HARD] [Water Resources — Open Channel Flow]

Prandtl's mixing length theory is used for:

A) Laminar flow
B) Turbulent flow
C) Compressible flow
D) Open channel

Q.35 [2 Marks] [HARD] [Environmental Engineering — Air Pollution]

CPM stands for:

A) Critical Path Method

B) Cost Per Mile

C) Continuous Process Model

D) Critical Process Method

Section B — Multiple Select Questions (MSQ)

Q.36 [2 Marks] [HARD] [Construction Management — Valuation]

Which are statically determinate?

A) Simply supported
B) Fixed-fixed
C) Cantilever
D) Continuous
E) Three-hinged arch

Q.37 [2 Marks] [HARD] [Structural Engineering — Influence Lines]

Which are statically determinate?

A) Simply supported
B) Fixed-fixed
C) Cantilever
D) Continuous
E) Three-hinged arch

Q.38 [2 Marks] [HARD] [Geotechnical Engineering — Slope Stability]

Which are statically determinate?

A) Simply supported
B) Fixed-fixed
C) Cantilever
D) Continuous
E) Three-hinged arch

Q.39 [2 Marks] [HARD] [Transportation Engineering — Traffic Engineering]

Which are statically determinate?

A) Simply supported
B) Fixed-fixed
C) Cantilever
D) Continuous
E) Three-hinged arch

Q.40 [2 Marks] [HARD] [Water Resources — Dams]

Which are statically determinate?

A) Simply supported
B) Fixed-fixed
C) Cantilever
D) Continuous
E) Three-hinged arch

Section C — Numerical Answer Type (NAT)

Q.51 [2 Marks] [HARD] [Transportation Engineering — Airport Planning]

Water flows at 2m/s in 200mm pipe. Reynolds number?

Answer: ____________________

Q.52 [2 Marks] [HARD] [Water Resources — Dams]

Sanitary landfill compaction target: density (kg/m^3)?

Answer: ____________________

Q.53 [2 Marks] [HARD] [Environmental Engineering — Water Treatment]

SS beam span 6m, UDL 10kN/m. Max BM (kN·m)?

Answer: ____________________

Q.54 [2 Marks] [HARD] [Construction Management — Contracts]

Rod A=500mm^2, P=50kN, E=200GPa, L=2m. Elongation (mm)?

Answer: ____________________

Q.55 [2 Marks] [HARD] [Structural Engineering — Deflection]

Water flows at 2m/s in 200mm pipe. Reynolds number?

Answer: ____________________

Q.56 [2 Marks] [HARD] [Geotechnical Engineering — Slope Stability]

Sanitary landfill compaction target: density (kg/m^3)?

Answer: ____________________

Q.57 [2 Marks] [HARD] [Transportation Engineering — Highway Planning]

SS beam span 6m, UDL 10kN/m. Max BM (kN·m)?

Answer: ____________________

Q.58 [2 Marks] [HARD] [Water Resources — Open Channel Flow]

Rod A=500mm^2, P=50kN, E=200GPa, L=2m. Elongation (mm)?

Answer: ______________________

Q.59 [2 Marks] [HARD] [Environmental Engineering — Noise Pollution]

Water flows at 2m/s in 200mm pipe. Reynolds number?

Answer: ____________________

Q.60 [2 Marks] [HARD] [Construction Management — Quality Control]

Sanitary landfill compaction target: density (kg/m³)?

Answer: ____________________

Answer Key

Verify your answers after completing the full paper.

Q.1:	B	Q.2:	A	Q.3:	B
Q.4:	B	Q.5:	A	Q.6:	B
Q.7:	A	Q.8:	B	Q.9:	B
Q.10:	A	Q.11:	B	Q.12:	A
Q.13:	B	Q.14:	B	Q.15:	A
Q.16:	B	Q.17:	A	Q.18:	B
Q.19:	B	Q.20:	A	Q.21:	B
Q.22:	A	Q.23:	B	Q.24:	B
Q.25:	A	Q.26:	B	Q.27:	A
Q.28:	B	Q.29:	B	Q.30:	A
Q.31:	B	Q.32:	A	Q.33:	B
Q.34:	B	Q.35:	A	Q.36:	ACE
Q.37:	ACE	Q.38:	ACE	Q.39:	ACE
Q.40:	ACE	Q.41:	ACE	Q.42:	ACE
Q.43:	ACE	Q.44:	ACE	Q.45:	ACE
Q.46:	ACE	Q.47:	ACE	Q.48:	ACE
Q.49:	ACE	Q.50:	ACE	Q.51:	400000
Q.52:	600	Q.53:	45	Q.54:	1
Q.55:	400000	Q.56:	600	Q.57:	45
Q.58:	1	Q.59:	400000	Q.60:	600

Detailed Solutions

Question 1

[Structural Engineering — Deflection]

Max BM in simply supported beam with UDL is at:

Answer: (B)

Max BM = $wL^2/8$ at midspan for SS beam with UDL.

Question 2

[Geotechnical Engineering — Soil Properties]

Terzaghi's bearing capacity assumes:

Answer: (A)

Terzaghi theory based on general shear failure mode.

Question 3

[Transportation Engineering — Traffic Engineering]

Bernoulli's equation is based on:

Answer: (B)

P/< r ^{2} $V^2/2g + z$ = constant. Conservation of energy.

Question 4

[Water Resources — Dams]

Prandtl's mixing length theory is used for:

Answer: (B)

Mixing length theory models turbulent shear stress.

Question 5

[Environmental Engineering — Water Treatment]

CPM stands for:

Answer: (A)

CPM = Critical Path Method for project scheduling.

Question 6

[Construction Management — Project Planning]

Max BM in simply supported beam with UDL is at:

Answer: (B)

Max BM = $wL^2/8$ at midspan for SS beam with UDL.

Question 7

[Structural Engineering — Influence Lines]

Terzaghi's bearing capacity assumes:

Answer: (A)

Terzaghi theory based on general shear failure mode.

Question 8

[Geotechnical Engineering — Slope Stability]

Bernoulli's equation is based on:

Answer: (B)

P/< r ^{2} $V^2/2g + z$ = constant. Conservation of energy.

Question 9

[Transportation Engineering — Pavement Design]

Prandtl's mixing length theory is used for:

Answer:

(B)

Mixing length theory models turbulent shear stress.

Question 10 *[Water Resources — Hydrology]*

CPM stands for:

Answer: (A)

CPM = Critical Path Method for project scheduling.

Question 11 *[Environmental Engineering — Noise Pollution]*

Max BM in simply supported beam with UDL is at:

Answer: (B)

Max BM = $wL^2/8$ at midspan for SS beam with UDL.

Question 12 *[Construction Management — CPM/PERT]*

Terzaghi's bearing capacity assumes:

Answer: (A)

Terzaghi theory based on general shear failure mode.

Question 13 *[Structural Engineering — Deflection]*

Bernoulli's equation is based on:

Answer: (B)

P/< r ^{2} $V^2/2g + z$ = constant. Conservation of energy.

Question 14 *[Geotechnical Engineering — Slope Stability]*

Prandtl's mixing length theory is used for:

Answer: (B)

Mixing length theory models turbulent shear stress.

Question 15 *[Transportation Engineering — Traffic Engineering]*

CPM stands for:

Answer: (A)

CPM = Critical Path Method for project scheduling.

Question 16 *[Water Resources — Dams]*

Max BM in simply supported beam with UDL is at:

Answer: (B)

Max BM = $wL^2/8$ at midspan for SS beam with UDL.

Question 17 *[Environmental Engineering — Sewage Treatment]*

Terzaghi's bearing capacity assumes:

Answer: (A)

Terzaghi theory based on general shear failure mode.

Question 18 *[Construction Management — Contracts]*

Bernoulli's equation is based on:

Answer:

(B)

P/< r ^{2} $V^2/2g$ + z = constant. Conservation of energy.

Question 19 *[Structural Engineering — Deflection]*

Prandtl's mixing length theory is used for:

Answer: (B)

Mixing length theory models turbulent shear stress.

Question 20 *[Geotechnical Engineering — Slope Stability]*

CPM stands for:

Answer: (A)

CPM = Critical Path Method for project scheduling.

Question 21 *[Transportation Engineering — Pavement Design]*

Max BM in simply supported beam with UDL is at:

Answer: (B)

Max BM = $wL^2/8$ at midspan for SS beam with UDL.

Question 22 *[Water Resources — Hydrology]*

Terzaghi's bearing capacity assumes:

Answer: (A)

Terzaghi theory based on general shear failure mode.

Question 23 *[Environmental Engineering — Noise Pollution]*

Bernoulli's equation is based on:

Answer: (B)

P/< r ^{2} $V^2/2g$ + z = constant. Conservation of energy.

Question 24 *[Construction Management — Valuation]*

Prandtl's mixing length theory is used for:

Answer: (B)

Mixing length theory models turbulent shear stress.

Question 25 *[Structural Engineering — Moment Distribution]*

CPM stands for:

Answer: (A)

CPM = Critical Path Method for project scheduling.

Question 26 *[Geotechnical Engineering — Slope Stability]*

Max BM in simply supported beam with UDL is at:

Answer: (B)

Max BM = $wL^2/8$ at midspan for SS beam with UDL.

Question 27 *[Transportation Engineering — Traffic Engineering]*

Terzaghi's bearing capacity assumes:

Answer:

(A)

Terzaghi theory based on general shear failure mode.

Question 28 *[Water Resources — Irrigation]*

Bernoulli's equation is based on:

Answer: (B)

P/< r ^{2} $V^2/2g$ + z = constant. Conservation of energy.

Question 29 *[Environmental Engineering — Noise Pollution]*

Prandtl's mixing length theory is used for:

Answer: (B)

Mixing length theory models turbulent shear stress.

Question 30 *[Construction Management — Quality Control]*

CPM stands for:

Answer: (A)

CPM = Critical Path Method for project scheduling.

Question 31 *[Structural Engineering — Deflection]*

Max BM in simply supported beam with UDL is at:

Answer: (B)

Max BM = $wL^2/8$ at midspan for SS beam with UDL.

Question 32 *[Geotechnical Engineering — Slope Stability]*

Terzaghi's bearing capacity assumes:

Answer: (A)

Terzaghi theory based on general shear failure mode.

Question 33 *[Transportation Engineering — Traffic Engineering]*

Bernoulli's equation is based on:

Answer: (B)

P/< r ^{2} $V^2/2g$ + z = constant. Conservation of energy.

Question 34 *[Water Resources — Open Channel Flow]*

Prandtl's mixing length theory is used for:

Answer: (B)

Mixing length theory models turbulent shear stress.

Question 35 *[Environmental Engineering — Air Pollution]*

CPM stands for:

Answer: (A)

CPM = Critical Path Method for project scheduling.

Question 36 *[Construction Management — Valuation]*

Which are statically determinate?

Answer:

(A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 37 *[Structural Engineering — Influence Lines]*

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 38 *[Geotechnical Engineering — Slope Stability]*

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 39 *[Transportation Engineering — Traffic Engineering]*

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 40 *[Water Resources — Dams]*

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 41 *[Environmental Engineering — Noise Pollution]*

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 42 *[Construction Management — Contracts]*

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 43 *[Structural Engineering — Determinate/Indeterminate]*

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 44 *[Geotechnical Engineering — Permeability]*

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 45 *[Transportation Engineering — Railways]*

Which are statically determinate?

Answer:

(A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 46 *[Water Resources — Hydraulics]*

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 47 *[Environmental Engineering — Air Pollution]*

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 48 *[Construction Management — CPM/PERT]*

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 49 *[Structural Engineering — Determinate/Indeterminate]*

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 50 *[Geotechnical Engineering — Bearing Capacity]*

Which are statically determinate?

Answer: (A, C, E)

Statically determinate: reactions from equilibrium alone.

Question 51 *[Transportation Engineering — Airport Planning]*

Water flows at 2m/s in 200mm pipe. Reynolds number?

Answer: (400000)

Re = VD/;Ò Ò -s ã"ó s ² b Ò C ‡GW& ulent).

Question 52 *[Water Resources — Dams]*

Sanitary landfill compaction target: density (kg/m³)?

Answer: (600)

Typical compacted MSW density "H 600 kg/m³.

Question 53 *[Environmental Engineering — Water Treatment]*

SS beam span 6m, UDL 10kN/m. Max BM (kN·m)?

Answer: (45)

Max BM = $wL^2/8 = 10\times36/8 = 45$ kN·m.

Question 54 *[Construction Management — Contracts]*

Rod A=500mm², P=50kN, E=200GPa, L=2m. Elongation (mm)?

Answer:

(1)

;B Ò Âô R Ò S s"òƒS s ² ms# s " Ò ã Ò Ò ÖÒà

Question 55 *[Structural Engineering — Deflection]*

Water flows at 2m/s in 200mm pipe. Reynolds number?

Answer: (400000)

Re = VD/;Ò Ò -s ã"ó s ² b Ò C ‡GW& ulent).

Question 56 *[Geotechnical Engineering — Slope Stability]*

Sanitary landfill compaction target: density (kg/m^3)?

Answer: (600)

Typical compacted MSW density "H 600 kg/m^3.

Question 57 *[Transportation Engineering — Highway Planning]*

SS beam span 6m, UDL 10kN/m. Max BM (kN·m)?

Answer: (45)

Max BM = $wL^2/8$ = 10×36/8 = 45 kN·m.

Question 58 *[Water Resources — Open Channel Flow]*

Rod A=500mm^2, P=50kN, E=200GPa, L=2m. Elongation (mm)?

Answer: (1)

;B Ò Âô R Ò S s"òƒS s ² ms# s " Ò ã Ò Ò ÖÒà

Question 59 *[Environmental Engineering — Noise Pollution]*

Water flows at 2m/s in 200mm pipe. Reynolds number?

Answer: (400000)

Re = VD/;Ò Ò -s ã"ó s ² b Ò C ‡GW& ulent).

Question 60 *[Construction Management — Quality Control]*

Sanitary landfill compaction target: density (kg/m^3)?

Answer: (600)

Typical compacted MSW density "H 600 kg/m^3.
